

Emergent Foraging, Architecture Competition, and World-Model Planning in a Multi-Agent Evolutionary Sandbox

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ABSTRACT

We study populations of neural agents that forage, learn within their lifetime by reinforcement learning, and reproduce with inheritance of trained weights in a procedurally generated 2-D world, and we report measured findings for three implemented capabilities. **First**, when a legacy recurrent brain and an attention-based brain *compete in one shared world* with offspring breeding true, the attention brain attains higher mean fitness (10.57 vs 8.95) and longer maximum lifespan (493 vs 259 ticks) and grows from one of eight founders to 96% of the population by tick 4000 — natural selection between architectures, in silico. **Second**, an observation-space world model $f(o,a) \rightarrow (\Delta o, \hat{r}, done)$ trained on 291k logged transitions predicts the next-state change 20% better than a static-world baseline (held-out Δo MSE 0.0268 vs 0.0337), explains $\approx 67\%$ of reward variance, and predicts termination at 0.999 accuracy. **Third**, equipping each agent with a per-genome latent dynamics model used for receding-horizon *planning* and prediction-error *curiosity* produces a behavioural phase change in a matched A/B: per-agent exploration nearly doubles (27.7 $\rightarrow$ 54.9 tiles), peak fitness rises 49% (38.7 $\rightarrow$ 57.8), and behaviour shifts from aimless turning (51% $\rightarrow$ 36% of actions) to the forage $\rightarrow$ eat $\rightarrow$ plant loop (eating $\times 6.5$, planting $\times 4$) — even though the model is only marginally better than chance at prediction. A population-size sweep shows the ecology self-regulates to a carrying capacity independent of founder count. The design adapts World Models [1], Dreamer/PlaNet latent imagination [2,3,4], random-shooting MPC [5], PPO [9] with GAE [10], and intrinsic-curiosity exploration [12]; our realizations are intentionally minimal. We report honest statistical caveats and release all configs, data, and code.

Keywords: multi-agent reinforcement learning · world models · model-based planning · intrinsic motivation · open-ended evolution · emergent behaviour

1. Introduction

Reinforcement-learning agents are usually trained one policy at a time against a fixed task. This work instead studies *populations* of agents that live, learn during their lifetime, and reproduce with inheritance of their learned weights (a Lamarckian scheme), so that selection and individual learning act together and behaviour can be *observed* rather than rewarded into existence. The substrate (Sec. 3) is a tile world with terrain and biomes, plants that grow and fruit, seeds that germinate, day/night and weather, optional fire, and an optional pheromone field.

This report does not introduce a new algorithm. It documents and quantitatively measures what has been built, establishing a reproducible baseline. Four findings emerge:

1. **Architectures can be selected ecologically.** With two brain designs in one world and breed-true reproduction, the stronger architecture wins by out-surviving and out-reproducing the other — no explicit benchmark (Sec. 6.1).
2. **A weak world model is still useful.** A model only 20% better than a "nothing changes" predictor still drives large control and exploration gains (Sec. 6.2–6.3).

3. Planning + curiosity trigger a behavioural phase change.

An online per-agent model flips the population from wandering to goal-directed foraging and a convergent shared strategy (Sec. 6.3).

4. **The ecology self-regulates.** From very different founder counts the population converges to the same carrying capacity (Sec. 6.4).

2. Related work

World models and latent imagination. Ha and Schmidhuber [1] showed agents can be trained inside a learned generative model; PlaNet [3] and the Dreamer line [2,4] learn latent dynamics and optimise behaviour by imagining rollouts. Our offline population model (Sec. 4.3) is the [1]-style predictor; the online per-agent latent head (Sec. 4.4) is the Dreamer-style component the planner and curiosity consume. We use the simplest instances — a two-layer MLP and random-shooting control — not a stochastic RSSM or a learned actor.

Decision-time planning. Random shooting and PETS [5] act on the best of many sampled model rollouts; MPPI [6] reweights trajectories; MuZero [7] runs value/policy-guided search over a learned model; MBPO [8] uses short model rollouts as extra training data. Our controller is the simplest of these and the clearest target for improvement.

Policy learning and curiosity. Lifetime learning uses PPO [9] with GAE(λ) [10] (a legacy A2C/A3C [11] learner exists too); intrinsic reward is forward-model prediction error, the ICM signal [12] (related: RND [13]). *Perception, memory, evolution.* The attention brain pools tokenised vision with one attention head [14] over a GRU [15]; reproduction is fission-with-mutation in the spirit of neuroevolution [16], and our behavioural-diversity metric follows novelty search [17].

3. The world: generation and dynamics

Each world is procedurally generated (Fig. 1). Layered value noise produces an elevation field; rivers are traced downhill

from peaks into basins; biomes (soil, sand, water, rock) follow elevation and a geography-driven moisture field, with fertile corridors along rivers. The ecology is then seeded: plants grow and, once mature, drop berries (food) and seeds; seeds germinate on fertile, moist, plantable tiles subject to a neighbourhood carrying-capacity cap. Per tick the world advances day/night and seasons, weather (rain/drought), an optional wildfire system on hot/dry vegetation, and slow soil fertility/moisture dynamics. Agents are then spawned on passable land. Everything is config-driven; a read-only bridge streams the live state to the viewers in Sec. 5 without affecting dynamics.

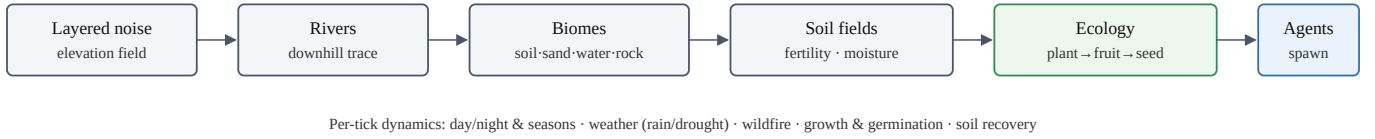


FIGURE 1. Procedural world-generation pipeline and the per-tick ecological dynamics layered on top.

4. Agent architecture and learning

Agents act once per tick on an egocentric, facing-rotated 5×5 vision grid (25 tiles × 2 channels = 50 features) plus internal

state, pre-digested stimulus features, inventory, and (v3.5) social/climate features. Three architectures share one genome/IO contract (Table 1); the genome layout is *append-only*, so a v3 genome migrates into v3.5 with bit-identical behaviour on the original actions.

TABLE 1. The three agent architectures (shared append-only genome/IO contract).

Brain	Obs	Actions	Params	Core
v2 (legacy)	72	8	≈8.9k	dense encoder → GRU → policy/value
v3 (attention)	72	8	≈17.3k	tokenised vision + attention → GRU → [z,h] value
v3.5 (social)	78	9	≈21.1k	v3 + 6 social/climate inputs + SIGNAL + pheromones

4.1 The v3 / v3.5 brain

Figure 2 shows the data flow. Each of the 25 vision tiles is embedded by one shared matrix (position-equivariant perception that scales with vision radius at constant parameter

count); a single attention query derived from the agent's internal state pools the tiles, so the network can focus on relevant cells (e.g. food when hungry) [14]. The latent $z=[state \mid pooled\text{-}vision]$ feeds a GRU [15] memory h ; the policy head reads h and the value head reads the joint $[z,h]$.

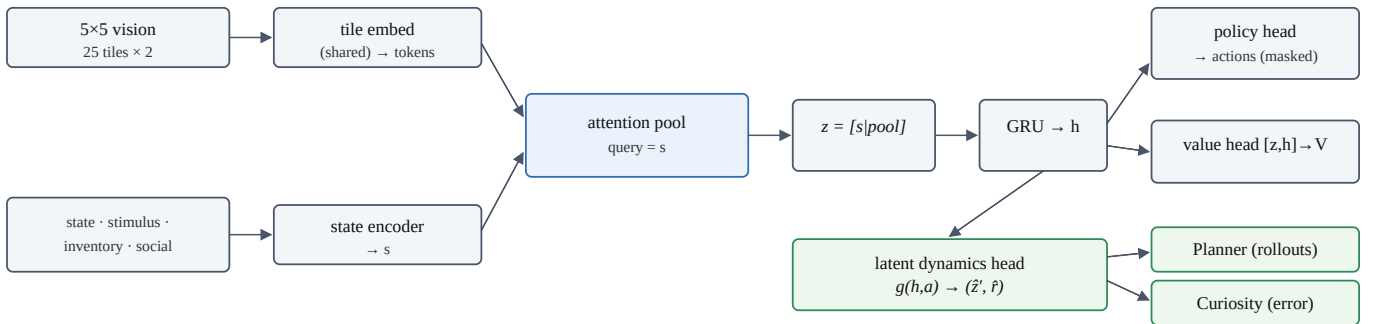


FIGURE 2. The v3/v3.5 agent brain. Green blocks are the optional latent world model and its two consumers (planner, curiosity), evaluated in Sec. 6.3.

4.2 Lifetime learning and inheritance

Agents learn during life with PPO [9] (clipped objective, $GAE(\lambda)$ [10], Adam, gradient clipping, GRU sequence replay); learning is globally budgeted per tick to bound wall-clock. At reproduction the child inherits the parent's *trained* weights with Gaussian mutation, so within-life learning is passed on and then shaped by selection [16].

4.3 Offline population world model

A shared, policy-agnostic model $f(o, a) \rightarrow (\Delta o, \hat{r}, p_{done})$ (two tanh layers, 128 hidden) is trained by regression on logged transitions (MSE on the observation delta and reward, BCE on termination), as in [1]. Predicting the *delta* concentrates capacity on what changes in a mostly-static world; it is the substrate for dream-based evolution.

4.4 Online planning and curiosity

Optionally each genome carries the latent dynamics head $g(h, a) \rightarrow (\hat{z}', \hat{r})$, trained as a PPO auxiliary loss. A *planner* does

receding-horizon random-shooting control [5]: each tick it imagines S rollouts of depth D in latent space, scores them $\sum_k \gamma^k \hat{r}_k + \gamma^D V$, and executes the first action of the best rollout, replanning next tick. A *curiosity* module adds the normalised forward-model prediction error as intrinsic reward [12].

5. Visualization: 2-D, isometric, and voxel views

The same live world can be observed three ways (Fig. 3): a top-down 2-D map (biome + elevation shading with entities), an isometric 2.5-D view (elevation as stacked tiles), and an interactive in-browser *voxel* renderer streamed over server-sent events (the read-only state bridge feeds all three). The voxel client shows terrain columns, trees that scale with maturity, berries, seeds, agents, weather, and a live HUD; its brain field reports the population's architecture mix (here $v3 \cdot 83 + v2 \cdot 17$, i.e. a $v3/v2$ cohort world).

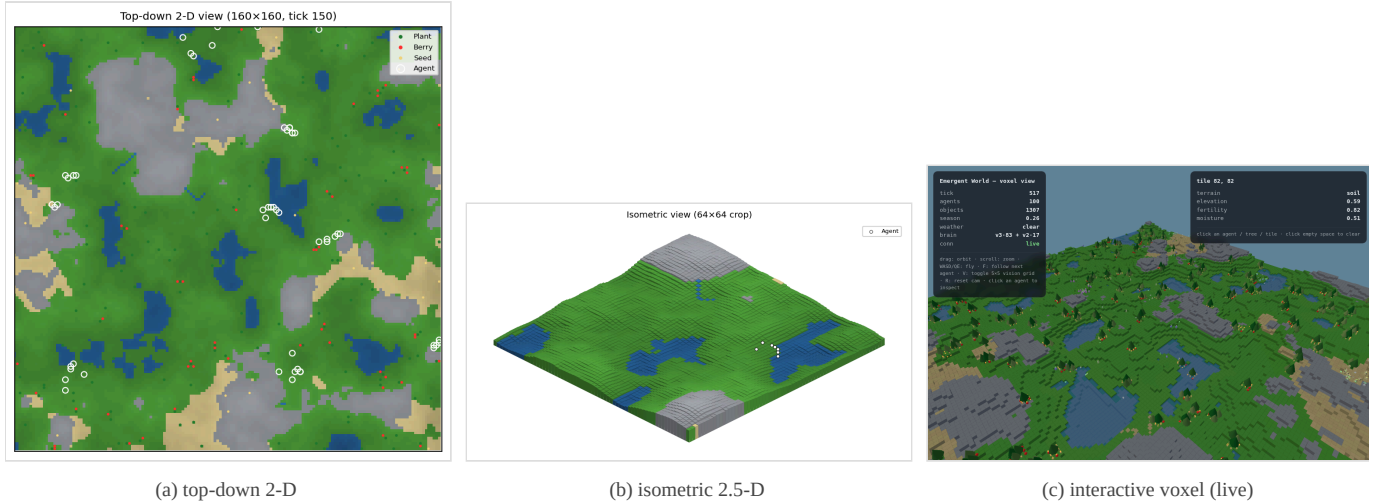


FIGURE 3. Three views of the *same* simulation, all from the read-only state bridge: (a) data-driven top-down map (160×160), (b) isometric render of a 64×64 crop, (c) the live browser voxel client.

6. Experiments and results

All runs are headless and config-driven; artifacts are committed under `docs/sample_*`. "Fitness" is the simulation's internal proxy and comparisons are within-simulation. Commands are in Sec. 9.

6.1 Ecological competition between architectures

In one 160×160 world, eight founders are seeded ≈15% legacy ($v2$) and the rest attention ($v3$); offspring breed true; 4000 ticks; cap 100. The attention brain dominates every survival/fitness proxy (Table 2) and population share moves from one $v2$ founder of eight to $\{v3:96, v2:4\}$. Both reach 100% eating success, so the edge is foraging efficiency and reproduction, not basic feeding.

TABLE 2. Cohort comparison from a single shared 4000-tick run (lifetime totals).

Cohort	Agents	Mean age	Max age	Mean fitness	Eat succ.
$v2$ (legacy)	255	98.3	259	8.95	100%
$v3$ (attention)	1715	101.3	493	10.57	100%

6.2 Accuracy of the learned world model

From 291,300 transitions logged during a v3.5+PPO run (64×64, population 30; all nine actions exercised, SIGNAL 67,027 times) the model trains 20 epochs (loss 3.20 → 1.71;

Fig. 4) and, on a 10% held-out split, beats the trivial baseline (Table 3). The absolute Δo gain is modest — expected for a two-layer deterministic model on a partly stochastic world — yet decisively useful downstream (Sec. 6.3).

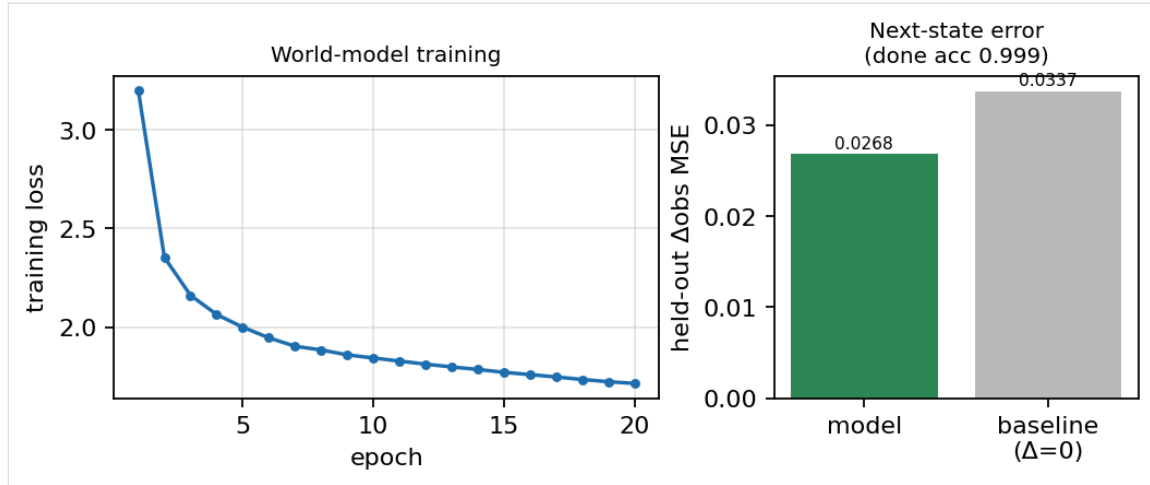


FIGURE 4. World-model training loss (left) and held-out next-state error vs. the predict- $\Delta=0$ baseline (right).

TABLE 3. Held-out predictive accuracy of the population world model.

Metric	Model	Baseline / scale	Reading
Δo MSE	0.0268	0.0337 ($\Delta=0$)	20% lower error
Reward MSE	1.70	variance 5.15	≈67% variance explained
done acc.	0.999	—	termination solved

6.3 Planning + curiosity: a behavioural phase change

Two runs are identical in seed (42), world (64×64), brain (v3.5), learner (PPO), and length (3000 ticks); the sole difference is whether the per-agent dynamics head, planner, and curiosity are

enabled. The effect is large across competence, exploration, and the action mix (Table 4). Per-agent strategy entropy *rises* while *pairwise* behavioural novelty *falls* (0.320 → 0.092): agents independently find the *same* effective strategy — the signature of a useful shared model, not noise [17].

TABLE 4. Matched A/B (single seed): planning + curiosity vs. baseline, v3.5+PPO, 3000 ticks.

Metric	Baseline	+Plan&Curiosity	Δ
Avg peak fitness	38.7	57.8	+49%
Mean fitness (final)	40.8	70.7	+73%
Avg lifespan (ticks)	422	548	+30%
Mean energy (final)	109	165	+51%
Tiles explored / agent	27.7	54.9	+98%
EAT attempts	185	1196	×6.5
Seeds planted	226	929	×4.1
Turning (L+R) share	51%	36%	−15pt
Strategy entropy	1.61	2.44	+0.83
Pairwise novelty	0.320	0.092	converges

6.4 Population dynamics across founder counts

To probe stability we ran three v3.5+PPO worlds (48×48, cap

60) seeded with 8, 24, and 48 founders (distinct seeds). Regardless of the starting count the population converges to the

same carrying capacity within a few hundred ticks and mean fitness rises over the run (Fig. 5): the ecology self-regulates

rather than exploding or collapsing — a prerequisite for the long-horizon studies above.

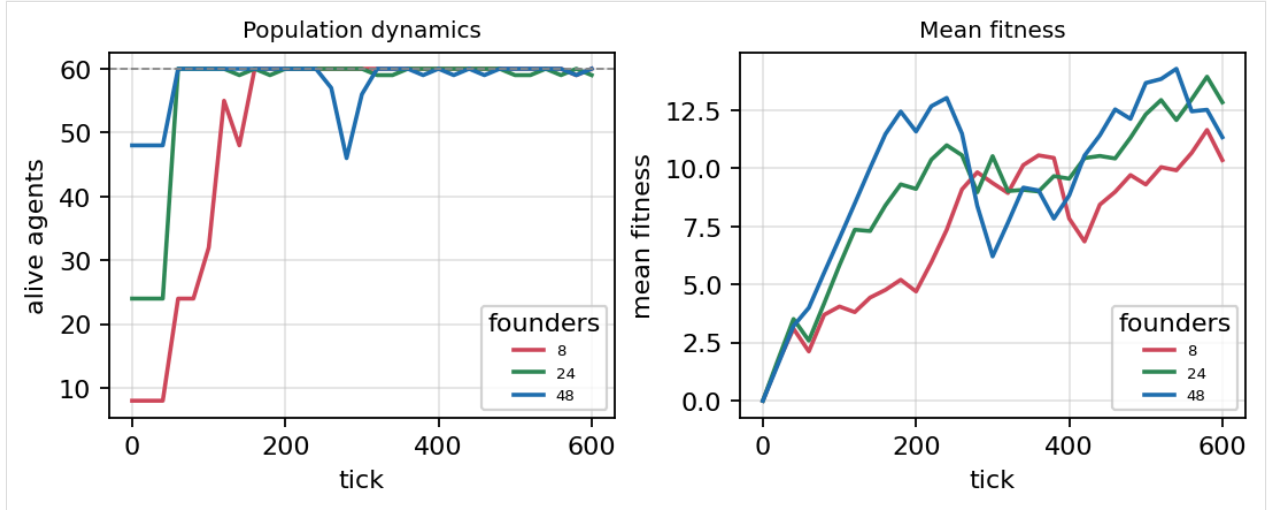


FIGURE 5. Population (left) and mean fitness (right) vs. tick for three founder counts. Different starts, same carrying capacity.

7. Discussion

The central, slightly counter-intuitive result is that a model only 20% better than a trivial baseline at next-state prediction still yields a $\sim 50\%$ fitness gain and a $\sim 2\times$ exploration gain once wired into planning and curiosity. Usefulness for control and exploration does not require predictive accuracy [1,2] — the model only has to rank actions and flag surprise, not reconstruct the world. Equally notable, the planner producing these gains is the weakest reasonable one (random shooting with uniform-random continuations, acting one step then replanning), so the effect should compound with better search (policy-guided imagination [2], CEM/MPPI [6], model-error discipline [5,8]).

8. Limitations

We state these so magnitudes are not over-read. The world-model and A/B studies (Sec. 6.2–6.3) are **single-seed**; Sec. 6.1 is one shared run with few founders — we report effect sizes, not significance, and multi-seed replication is the obvious next step. Horizons are short (2k–10k ticks) on small worlds (48–160 per side, 30–100 agents), bounded by the $\sim 1\text{--}8$ ticks/s cost of v3.5+PPO+planning. Fitness is an in-silico proxy. The

dynamics models are deterministic MLPs and the planner ignores uncertainty. Sec. 6.3 toggles planner and curiosity together; separating them is a planned ablation.

9. Conclusion

In one multi-agent sandbox we observed architecture selection by ecological competition, trained a compact world model that beats a trivial baseline, showed that online planning plus curiosity built on such a model produces a clear shift to goal-directed, convergent foraging, and verified that the ecology self-regulates to a carrying capacity independent of founder count. The results are modest in scale but reproducible end-to-end, and they establish the baseline against which the staged planner upgrades (policy-guided imagination, CEM, and an imagination-trained actor) can be measured. All code, configs, run artifacts, and the three viewers are released. Reproducibility: Sec. 6.1 `scripts/competition_run.py`; Sec. 6.2 `main.py --world-model-log` then `scripts/train_world_model.py`; Sec. 6.3 `two main.py --seed 42` runs (`worldmodel_v35.yaml` vs `planning_curiosity_v35.yaml`) compared with `scripts/analyze_logs.py`; Sec. 6.4 a three-config population sweep. Figures regenerate via `paper/make_figures.py`.

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This report describes only implemented, measured functionality of the Emergent World-Model Sandbox as of commit e0551bd. Figures are generated from real simulation state and committed run artifacts. Cited works are the sources of ideas adapted here; the implementations are deliberately simplified and any shortfalls are the author's.